Linear System Theory

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• The characteristic polynomial of A:

$$\text{charpoly}(A) := \det(sI - A)$$

Note that $\det(sI - A)$ is an n-th order monic polynomial, that is, it has the form

$$\det(sI - A) = a_0 + a_1 s + \ldots + s^n$$

• We conclude that a complex number λ is an eigenvalue of A if and only if it is a root of the nth order characteristic polynomial of A. Hence, A has at least one eigenvalue and at most n distinct eigenvalues. Suppose A has l distinct eigenvalues, $\lambda_1, \lambda_2, \ldots, \lambda_l$; since these are the distinct roots of the characteristic polynomial of A, we must have

$$\det(sI - A) = \prod_{i=1}^{l} (s - \lambda_i)^{m_i}$$

• The integer m_i is called the algebraic multiplicity of λ_i

• It should be clear that if v^1 and v^2 are any two eigenvectors corresponding to the same eigenvalue λ and ξ_1 and ξ_2 are any two numbers then (provided it is nonzero) $\xi_1 v^1 + \xi_2 v^2$ is also an eigenvector for λ. The set of eigenvectors corresponding to λ along with the zero vector is called the eigenspace of A associated with λ. This is simply the null space of $\lambda I - A$

• The geometric multiplicity of λ is the nullity of $\lambda I - A$, that is, it is the maximum number of linearly independent eigenvectors associated with λ.

• It follows that A is invertible if and only if all its eigenvalues are nonzero.

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Companion matrix

$$p(s) = a_0 + a_1 s + a_2 s^2 + \cdots + a_{n-1} s^{n-1} + a_n s^n$$

$$\updownarrow \qquad\qquad \det(sI - A)$$

$$A = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & & & & & \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-2} & -a_{n-1} \end{pmatrix}$$

given a bunch of n complex numbers, there is a companion matrix whose eigenvalues are exactly those numbers.

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Behavior of continuous-time systems (LTI)

The system $\dot{x} = Ax$ has a non-zero solution of the form

$$x(t) = e^{\lambda t} v$$

if and only if λ is an eigenvector of A and v is a corresponding eigenvector.

eigenvalue

* $\lambda v = Av$

$x(t) = e^{\lambda t} v$

$\dot{x} = \lambda e^{\lambda t} v$

$= e^{\lambda t} \lambda v$

$= e^{\lambda t} Av$

$= Ax$

* $e^{\lambda t} v$ is a very special type of sol

"mode" of system

* $x(0) = v$

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If v is an eigenvector then, $e^{\lambda t}v$ is also an eigenvector for each t. Also, considering the magnitude of $e^{\lambda t}v$, we obtain

scalar

$\lambda \in \mathbb{C} \rightarrow \alpha + j\omega$ ↳ oscillation

scalar

$$\|e^{\lambda t}v\| = |e^{\lambda t}|\,\|v\|$$

$$= e^{\alpha t}\,\|v\| \qquad \alpha = \Re(\lambda)$$

(a) Once an eigenvector, always an eigenvector

(b) If $\alpha < 0$, you decay.

(c) If $\alpha > 0$, you grow.

(d) If $\alpha = 0$, your magnitude remains constant.

(e) If $\lambda = 0$, $\exp(0) = 1$, every eigenvector corresponding to $\lambda = 0$ is an eq. state

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Example 62 (An eigenvalue-eigenvector calculation)

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

$\det(\lambda I - A) = \det\begin{pmatrix} \lambda-3 & 1 \\ 1 & \lambda-3 \end{pmatrix}$

$= (\lambda-3)^2 - 1$

$= \lambda^2 - 6\lambda + 8$

$= (\lambda-2)(\lambda-4)$

$\lambda = 2, \lambda = 4$

$AM = 1 \quad AM = 1$

$(\lambda I - A)\cdot v = 0$

$\lambda v = Av$ $\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 5 \\ 5 \end{pmatrix} = \begin{pmatrix} 10 \\ 10 \end{pmatrix} = 2\begin{pmatrix} 5 \\ 5 \end{pmatrix}$

① $\lambda = 2$

$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} v = 0 \quad v = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$

② $\lambda = 4$

$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} v = 0 \quad v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$Av = 2v$

$\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 2\cdot\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

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Example 66 The matrix A of example 62 had 2 as an eigenvalue with corresponding eigenvector $[\ 1 \quad 1\ ]^T$. Hence with initial state,

$\dot{X} = AX$ $\qquad X_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $\qquad X_0 = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$

the system has following sol:

$X(t) = e^{\lambda t} \cdot V = e^{2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} e^{2t} \\ e^{2t} \end{pmatrix}$

(X_0)

$\begin{pmatrix} 5 \\ 5 \end{pmatrix} = \begin{pmatrix} 5e^{2t} \\ 5e^{2t} \end{pmatrix}$

$\lambda = 4, \quad V = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\Rightarrow X_0 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad X(t) = e^{4t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} e^{4t} \\ -e^{-4t} \end{pmatrix}$

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The behaviour of the sys in the 1D subspace spanned by v is that of a scalar sys

Behaviour in the v subspace

Suppose v is an e-vector of $A \Leftrightarrow$ e-value λ

Consider any initial value $x_0 = \underbrace{\xi_0}_{\text{scalar}} v$

$= \text{span}\{v\}$

$x(t) = \xi(t) v$

$\dot{x} = \dot{\xi} v = Ax = A(\xi v) = \lambda \xi v$

$\boxed{\dot{\xi} = \lambda \xi}$, $\xi(0) = \xi_0$

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$\lambda = 1 + j \iff \bar{\lambda} = 1 - j$ the sol depends linearly on x_0

$x(t) = e^{\lambda t} \cdot x_0$

Real systems and complex eigenvalues

- Since A is real and λ is complex, the eigenvector v is complex. The complex system can start with a complex initial state but the 'real' physical system cannot.

$\bar{\lambda}$ is also an eigenvalue of A w/ corresponding eigenvector $\bar{v}$, $x(0) = \bar{v}$

$x(t) = e^{\bar{\lambda} t} \cdot \bar{v} = \overline{e^{\lambda t} v}$

Consider real vector

$u := \Re(v) = \frac{1}{2}(v + \bar{v})$ as initial state

$x(t) = \frac{1}{2}(e^{\lambda t} v + \overline{e^{\lambda t} v}) = \Re(e^{\lambda t} v)$

$\alpha = \Re(\lambda)$ $\omega = \Im(\lambda)$, $u = \Re(v)$, $p = \Im(v)$

$\lambda = \alpha + j\omega$ $\quad$ $v = u + jp$

$$e^{\lambda t} v = e^{\alpha t}(\cos \omega t + j \sin \omega t)(u + jp)$$

$$= e^{\alpha t}(\cos(\omega t) \cdot u - \sin(\omega t) \cdot p) \quad R$$

$$+ j\left(e^{\alpha t}(\sin(\omega t) \cdot u + \cos(\omega t) \cdot p)\right) \quad I$$

$\xi_1(t)u$

$x(0) = u \Rightarrow x(t) = e^{\alpha t}(\cos(\omega t) \cdot u - \sin(\omega t) \cdot p)$ $\neq \Re(e^{\lambda t} v)$

$x(0) = p = \frac{1}{2j}(e^{\lambda t} v - e^{\overline{\lambda} t} \overline{v})$

$x(t) = \left(e^{\alpha t}(\sin(\omega t) \cdot u + \cos(\omega t) \cdot p)\right)$

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- So, if the initial state is of the form

$$x(0) = \xi_{10}u + \xi_{20}w$$

where ξ_{10}, ξ_{20} are any two real scalars (that is, the system starts in the real subspace spanned by the real span of u, w), the resulting solution is given by

$$x(t) = \xi_1(t)u + \xi_2(t)w$$

scalar

$x(t) = 0\,u + 0\,p$

$x(t) = 0\,u + 0\,p$

$$\xi_1(t) = e^{\alpha t}(\cos(\omega t)\xi_{10} + \sin(\omega t)\xi_{20})$$

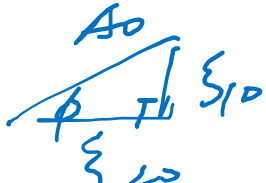
$$\xi_2(t) = -e^{\alpha t}(\sin(\omega t)\xi_{10} + \cos(\omega t)\xi_{20})$$

$$= \begin{pmatrix} e^{\alpha t}\cos(\omega t) & e^{\alpha t}\sin(\omega t) \\ -e^{\alpha t}\sin(\omega t) & e^{\alpha t}\cos(\omega t) \end{pmatrix} \begin{pmatrix} \xi_{10} \\ \xi_{20} \end{pmatrix}$$

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$$A_0 = (\xi_{10}^2 + \xi_{20}^2)^{1/2}, \quad \phi_0 \in [0, 2\pi)$$

$$\sin(\phi_0) = \xi_{10}/A_0 \qquad \cos(\phi_0) = \xi_{20}/A_0$$



$$\Rightarrow \begin{cases} \xi_1(t) = A_0 e^{\alpha t} \sin(\omega t + \phi_0) \\ \xi_2(t) = A_0 e^{\alpha t} \cos(\omega t + \phi_0) \end{cases}$$

$$\Rightarrow x(t) = A_0 e^{\alpha t} R(\lambda) \begin{pmatrix} \sin(\omega t + \phi_0) u + \\ \cos(\omega t + \phi_0) p \end{pmatrix}$$

$\operatorname{span}\{u, p\}$

* If the initial state is in the 2D subspace spanned by u, p, then so is the resulting $sol(x(t))$

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$A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ $\lambda = 1 \pm j$

$v = \begin{pmatrix} 1 \\ -j \end{pmatrix}, \begin{pmatrix} 1 \\ j \end{pmatrix}$

Example 67 Recall example 63. Considering

$$v = v^1 = \begin{pmatrix} 1 \\ j \end{pmatrix}, \lambda = 1 + j$$

real $u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ Img. $p \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\alpha = 1 = \Re(\lambda)$

$\omega = -1 = \Im(\lambda)$

scalar

$x_0 = x_{10} u + x_{20} p$

$x(t) = x_1(t) u + x_2(t) p$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} e^{\alpha t}\cos(\omega t) & e^{\alpha t}\sin(\omega t) \\ -e^{\alpha t}\sin(\omega t) & e^{\alpha t}\cos(\omega t) \end{bmatrix} \begin{bmatrix} x_{10} \\ x_{20} \end{bmatrix}$$

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Behavior in the $u-w$ subspace. We have shown that if the state starts in the 2-dimensional subspace spanned by u, w, (let's call it the $u-w$ subspace) it remains there, that is, if $x_0 = \xi_{10} u + \xi_{20} w$ then $x(t) = \xi_1(t) u + \xi_2(t) w$. Forgetting about the actual solution, we now show that motion in this subspace is described by:

$$\begin{bmatrix} \dot{\xi}_1 \\ \dot{\xi}_2 \end{bmatrix} = \begin{bmatrix} \alpha \xi_1 + \omega \xi_2 \\ -\omega \xi_2 + \alpha \xi_1 \end{bmatrix} = \begin{bmatrix} \alpha & \omega \\ -\omega & \alpha \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix}$$

$Av = \lambda v \Rightarrow A(u + jp) = (\alpha + j\omega)(u + jp)$

$$Au = \alpha u - \omega p$$

$$Ap = \omega u + \alpha p$$

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$$Au = \alpha u - \omega p$$

$$Ap = \omega u + \alpha p$$

$$x(t) = \xi_1(t)\,u + \xi_2(t)\,p$$

$$\begin{bmatrix} \dot{\xi}_1 \\ \dot{\xi}_2 \end{bmatrix} = \begin{bmatrix} \alpha & \omega \\ -\omega & \alpha \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix}$$

$$\dot{x} = \dot{\xi}_1 u + \dot{\xi}_2 p$$

$$= Ax$$

$$= A(\xi_1 u + \xi_2 p)$$

$$= \xi_1 Au + \xi_2 Ap$$

$$= \xi_1(\alpha u - \omega p) + \xi_2(\omega u + \alpha p)$$

$$= (\alpha\xi_1 + \omega\xi_2)u + (-\omega\xi_1 + \alpha\xi_2)p$$

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Example 68 (Beavis and Butthead) Consider $u_1 = u_2 = 0$ in the system in Figure 6.3.

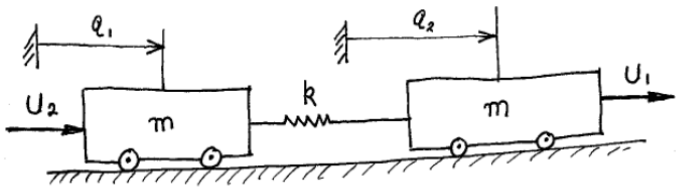


Figure 6.3: Beavis & Butthead

$$\begin{aligned} m\ddot{q}_1 &= k(q_2 - q_1) \\ m\ddot{q}_2 &= -k(q_2 - q_1) \end{aligned}$$

$x_1 = q_1 \quad x_2 = q_2$

$x_3 = \dot{q}_1 \quad x_4 = \dot{q}_2$

$\Rightarrow \dot{x} = Ax$

$$A = \begin{pmatrix} 0_{2\times 2} & I_{2\times 2} \\ A_{21} & 0_{2\times 2} \end{pmatrix}$$

$$A_{21} = k/m \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$$

$\lambda_1 = j\omega \,,\ \lambda_2 = -j\omega \,,\ \lambda_3 = \lambda_4 = 0$

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$\lambda_1 = j\omega$, $\lambda_2 = -j\omega$, $\lambda_3 = \lambda_4 = 0$

$v^1 = \begin{pmatrix} 1 \\ -1 \\ j\omega \\ -j\omega \end{pmatrix}$, $v^2 = \overline{v^1}$, $v^3 = v^4 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$

$S^1 = \text{span}\left\{ \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix} \right\} \Rightarrow q_2 = -q_1$

$\dot{q}_2 = -\dot{q}_1$

A state traj which remains in S^1 is characterized by the condition $q_2(t) \equiv -q_1(t)$

Every state in this subspace had the form $\begin{pmatrix} q_1 \\ \dot{q}_1 \\ 0 \\ 0 \end{pmatrix} \Rightarrow$ an eg state

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Solution of nondefective matrices

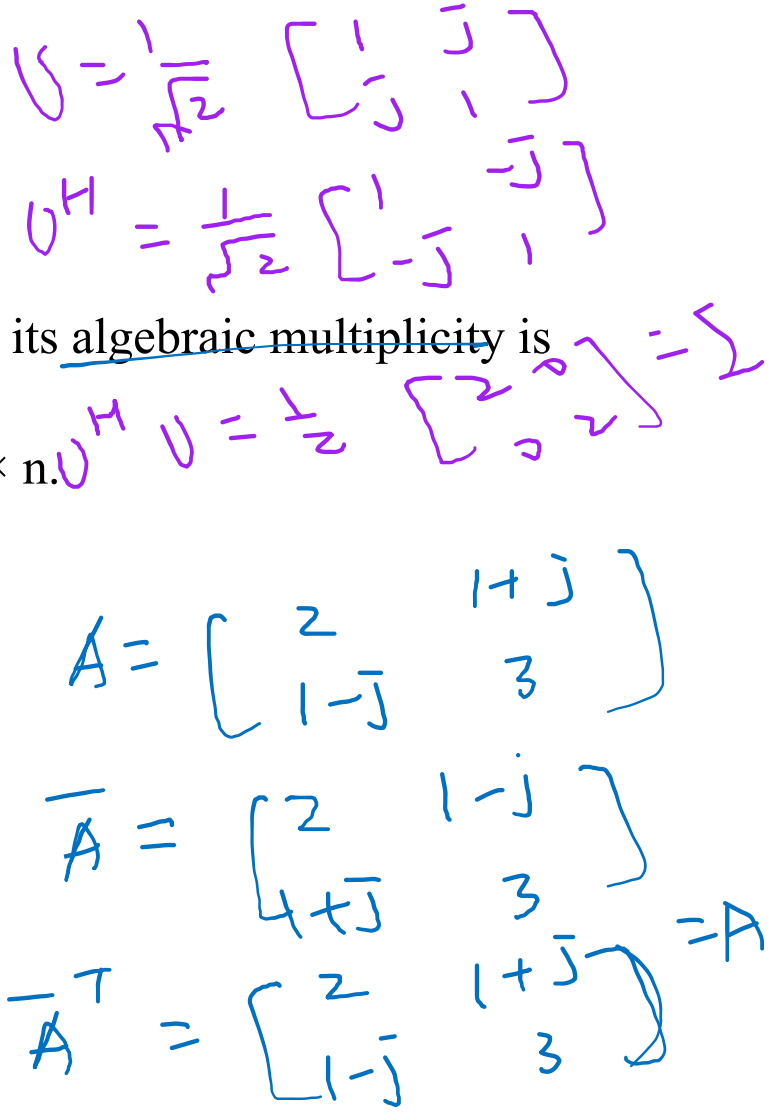
- An eigenvalue whose geometric multiplicity equals its algebraic multiplicity is call a **nondefective eigenvalue**

  (a) A has n distinct eigenvalues where A is n × n.

  (b) A is hermitian that is, $A^H = A$

  (c) A is unitary, that is, $A^H A = I$

Here A^H is the complex conjugate transpose of A.



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有 n 個 不一樣的 eigen value

Suppose A is a nondefective matrix and consider any initial condition: $x(0) = x_0$. Since A is nondefective, it has a basis of eigenvectors $v^1, \cdots, v^n$. Hence, there are unique scalars $\xi_{10}, \xi_{10}, \ldots, \xi_{n0}$ such that x_0 can be expressed as

scalar

ξ_{20}

$$x_0 = \xi_{10}v^1 + \xi_{20}v^2 + \ldots + \xi_{n0}v^n. \quad \in \text{span}\{v^1, v^2, \cdots, v^n\}$$

$x_0 = \xi_{10} v^1$

$x(t) = e^{\lambda_1 t} \xi_{10} v^1$

$x(t) = e^{\lambda_1 t}\xi_{10}v^1 + e^{\lambda_2 t}\xi_{20}v^2 + \cdots + e^{\lambda_n t}\xi_{n0}v^n$

"mode"

$e^{\lambda_i t}\xi_{i0}v^i \rightarrow$ "mode"

* Every motion (solution) of a LTI sys w/ nondefective A matrix is "a sum of modes"

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$(A-\lambda I)v = (\lambda I - A)v = 0$

Defective matrices and generalized eigenvectors

Fact 9 *An eigenvalue λ for A is defective if and only if there is a nonzero vector g such that*

$(A-\lambda I)^3 g = 0 \Rightarrow (A-\lambda I)g = v \neq 0$

$(A-\lambda I)(A-\lambda I)g = (A-\lambda I)v = 0$

g is called a "generalized eigenvector" for A

Jordan form (block)

$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ $\lambda = 0$. $v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} g = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$g = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

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We now demonstrate the consequence of a system having a defective eigenvalue. So suppose λ is a defective eigenvalue for a square matrix A. Let g be any corresponding generalized eigenvector for λ. We now show that

$$x(t) = e^{\lambda t} g + t e^{\lambda t} (A - \lambda I) g$$

$$\dot{x}(t) = e^{\lambda t} \lambda g + e^{\lambda t} (A - \lambda I) g + t e^{\lambda t} \lambda (A - \lambda I) g$$

$$= e^{\lambda t} A g + t e^{\lambda t} \lambda (A - \lambda I) g$$

$$\lambda (A - \lambda I) g = A (A - \lambda I) g$$

$$\dot{x}(t) = e^{\lambda t} A g + t e^{\lambda t} A (A - \lambda I) g$$

$$= A x$$

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Consider the function given by $te^{\lambda t}$ and note that

$$|te^{\lambda t}| = |t|e^{\alpha t}$$

$\lambda \in \mathbb{C}$ exp 會隨著 t 的絕對值 α 而改變

if $\alpha < 0 \Rightarrow$ decay exponentially

if $\alpha > 0 \Rightarrow$ grows exponentially

if $\alpha = 0 \Rightarrow |t|$ unbounded.

↳ ① $\lambda = 0$ // ② $\alpha = 0$ 代表為 imaginary part

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Example 72 (Resonance) Consider a simple spring-mass system, of mass m and spring constant k, subject to a sinusoidal force $w(t) = A\sin(\omega t + \phi)$ where $\omega = \sqrt{k/m}$ is the unforced natural frequency of the spring mass system and A and ϕ are arbitrary constants. Letting q be the displacement of the mass from its unforced equilibrium position, the motion of this system can be described by

$$m\ddot{q} + kq = w.$$

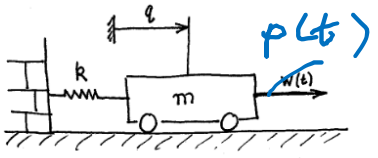


Figure 6.4: Resonance

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$\lambda_1 = j\omega$, $\lambda_2 = -j\omega$ $\quad AM = 2$

$v^1 = \begin{pmatrix} 1 \\ j\omega \\ 0 \\ 0 \end{pmatrix}$ $\quad v^2 = \begin{pmatrix} 1 \\ -j\omega \\ 0 \\ 0 \end{pmatrix}$ $\quad GM = 1$

$\hookrightarrow g^1 = \begin{pmatrix} \frac{1}{2}k \\ 0 \\ -j\omega \end{pmatrix}$ $\quad g^2 = \begin{pmatrix} \frac{1}{2}k \\ 0 \\ 1 \\ -j\omega \end{pmatrix}$

$(A-\lambda I)g = 0$ $\quad x(t) = e^{\lambda t} g = t e^{\lambda t}(A-\lambda I)g$

$\langle t\sin\omega t \,/\, t\cos\omega t$

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Behavior of discrete-time systems

Consider now a discrete time system described by

$$x(k+1) = Ax(k)$$

We now demonstrate the following result:

$x(k) = \lambda^k v$ — initial state

$x(k) = \lambda^k x_0$ x_0 可以寫 v

$\lambda, v \Rightarrow A$'s eigenvalue, eigenvector

$Av = \lambda v$

$x(k+1) = \lambda^{k+1} v = \lambda \cdot \lambda^k v$

$= A\lambda^k v = Ax(k)$

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A solution of the form $\lambda^k v$ is a very special type of solution and is sometimes called a mode of the system. Note that if $x(k) = \lambda^k v$ is a solution, then $x(0) = v$, that is, v is the initial value of x. Hence, we can make the following statement.

If v is an eigenvector of A, then the solution to $x(k+1) = Ax(k)$ with initial condition $x(0) = v$ is given by $x(k) = \lambda^k v$ where λ is the eigenvalue corresponding to v.

If v is an eigenvector then, $\lambda^k v$ is also an eigenvector for each k. Also, considering the magnitude of $\lambda^k v$, we obtain

$$\|\lambda^k v\| = |\lambda^k| \, \|v\| = |\lambda|^k \|v\|$$

$x(0) = v$

* $\lambda = 0$, $x(k) = 0$ for all $k > 0$ the state of sys goes to 0 in 1 step

(a) Once an eigenvector, always an evector!

(b) $|\lambda| < 1 \Rightarrow$ decay

(c) $|\lambda| > 1 \Rightarrow$ grow

(d) $|\lambda| = 1 \rightarrow \lambda = 1 \rightarrow$ v is of state remains the same

$\lambda = -1 \rightarrow$ oscillate!

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All solutions for nondefective matrices

- Recall that an eigenvalue whose geometric multiplicity equals its algebraic multiplicity is called a nondefective eigenvalue. Otherwise, it is considered defective.
- A matrix with the property that all its eigenvalues are nondefective is called a nondefective matrix. Otherwise, it is considered defective.
- Recall also that an $n \times n$ matrix has n linearly independent eigenvectors if and only if A is nondefective.

GM = AM

n個不一樣的 evalue

evector

n

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All solutions for nondefective matrices

Example 73 The following matrix is a simple example of a defective matrix.

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Suppose A is a nondefective matrix and consider any initial condition: $x(0) = x_0$. Since A is nondefective, it has a basis of eigenvectors $v^1, \cdots, v^n$. Hence,

$$x_0 = \xi_{10} v^1 + \xi_{20} v^2 + \ldots + \xi_{n0} v^n$$

$$x(k) = \lambda_1^k \xi_{10} v^1 + \lambda_2^k \xi_{20} v^2 + \cdots + \lambda^n \xi_{n0} v^n$$

mode

$\Rightarrow$ a sum of modes

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$x(k+1) = A\,x(k)$

Some solutions for defective matrices

We now demonstrate the consequence of a system having a defective eigenvalue. So, suppose λ is a defective eigenvalue for a square matrix A. Let g be any corresponding generalized eigenvector for λ. We now show that

$$x(k) = \lambda^k g + k\lambda^{k-1}(A-\lambda I)g$$

$$A(A-\lambda I)g = \lambda(A-\lambda I)g$$

$$Ax(k) = \lambda^k Ag + k\lambda^{k-1}A(A-\lambda I)g$$

$$= \lambda^k Ag + k\lambda^k(A-\lambda I)g$$

$$x(k+1) = \lambda^{k+1} g + (k+1)\lambda^k(A-\lambda I)g$$

$$= \lambda^{k+1} g + k\lambda^k(A-\lambda I)g + \lambda^k Ag$$

$$= \lambda^{k+1} g$$

$$x(k+1) = Ax(k)$$

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Some solutions for defective matrices

Consider the function given by $k\lambda^{k-1}$ and note that

$$|k\lambda^{k-1}| = |k||\lambda|^{k-1}.$$

① $|\lambda| < 1 \rightarrow$ decay

② $|\lambda| > 1 \rightarrow$ grow

③ $|\lambda| = 1 \rightarrow |k|$ unbounded

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Coordinate transformations. Suppose we have n scalar variables $x_1, x_2, \ldots, x_n$ and we implicitly define new scalar variables $\xi_1, \xi_2, \ldots, \xi_n$ by

$$\boxed{x = T\xi} \quad \text{where} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \text{and} \quad \xi = \begin{pmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_n \end{pmatrix}$$

and T is an $n \times n$ invertible matrix. Then, ξ is explicitly given by

$$\xi = T^{-1}x$$

We can obtain a geometric interpretation of this change of variables as follows. First, observe that

$$\underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{x} = x_1 \underbrace{\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{e^1} + x_2 \underbrace{\begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}}_{e^2} + \ldots + x_n \underbrace{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}}_{e^n},$$

that is the scalars $x_1, x_2, \ldots, x_n$ are the coordinates of the vector x wrt the standard basis $e^1, e^2, \ldots, e^n$, or, x is the coordinate vector of itself wrt the standard basis. Suppose

$$T = \begin{pmatrix} t^1 & \ldots & t^j & \ldots & t^n \end{pmatrix}$$

that is, t^j is the j-th column of T. Since T is invertible, its columns, $t^1, t^2, \ldots, t^n$, form a basis for $\mathbb{C}^n$. Also, $x = T\xi$ can be written as

$$x = \xi_1 t^1 + \xi_2 t^2 + \ldots + \xi_n t^n$$

From this we see that $\xi_1, \xi_2, \ldots, \xi_n$ are the coordinates of x wrt to the new basis and the vector ξ is the coordinate vector of the vector x wrt this new basis. So $x = T\xi$ defines a coordinate transformation.

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A square A matrix is said to be "similar" to another square matrix Λ if there exists T is nonsingular such that $\Lambda = T^{-1}AT$

Similarity transformations

Similarity transformations. Consider now a CT (DT) system described by

$$\dot{x} = Ax \qquad (x(k+1) = Ax(k))$$

and suppose we define new state variables by $x = T\xi$ where T is nonsingular; then the behavior of ξ is governed by

basis invertible

$$\dot{\xi} = \Lambda \xi \qquad \xi(k+1) = \Lambda \xi(k)$$

$n \times n$

$$\Rightarrow \Lambda = T^{-1} A T$$

$$\dot{x} = T\dot{\xi} = A(T\xi)$$

$$\dot{\xi} = \boxed{T^{-1}AT}\,\xi$$

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Useful result. Suppose

$$At^j = \alpha_{1j}t^1 + \alpha_{2j}t^2 + \ldots + \alpha_{nj}t^n$$

Then the matrix $\Lambda = T^{-1}AT$ is uniquely given by

$$\Lambda_{ij} = \alpha_{ij}$$

that is Λ_{ij} is the i-th coordinate of the vector At^j wrt the basis $t^1, t^2, \cdots, t^n$. Thus, *the j-th column of the matrix Λ is the coordinate vector of At^j wrt the basis $t^1, t^2, \cdots, t^n$.*

proof : $T\Lambda = AT$.

$$T\Lambda = (t^1, t^2, \cdots, t^n) \begin{pmatrix} \Lambda_{11} & \cdots & \Lambda_{1n} \\ \Lambda_{21} & & \\ \vdots & & \\ \Lambda_{n1} & \cdots & \Lambda_{nn} \end{pmatrix}$$

$$AT = A(t^1, t^2, \cdots, t^n) \; t^j : n\times 1$$

$$= (At^1, At^2, \cdots, At^n) = (\Lambda_{11}t^1 + \cdots + \Lambda_{n1}t^n \; \cdots \; \Lambda_{1n}t^1 + \cdots + \Lambda_{nn}t^n)$$

j-th col $= \Lambda_{1j}t^1 + \cdots + \Lambda_{nj}t^n$

$= A \cdot t^j$

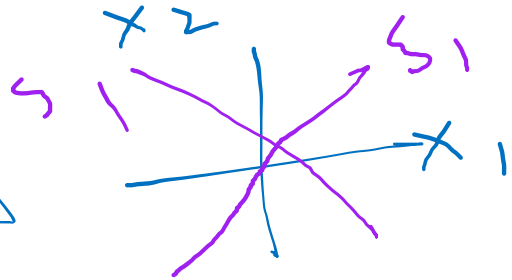
$n\times 1$

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Motivating example. Consider

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

$$\begin{aligned} \dot{x}_1 &= 3x_1 - x_2 \\ \dot{x}_2 &= -x_1 + 3x_2 \end{aligned}$$



Suppose we define new states:

$$\begin{aligned} \xi_1 &= (x_1 + x_2)/2 \\ \xi_2 &= (x_1 - x_2)/2 \end{aligned}$$

Then,

$$\begin{aligned} \dot{\xi}_1 &= 2\xi_1 \\ \dot{\xi}_2 &= 4\xi_2 \end{aligned}$$

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$x_1 = \xi_1 + \xi_2$

$x_2 = \xi_1 - \xi_2$

$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$

$x = x_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ +

$= \xi_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \xi_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ ×

$T = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ → $(v_1 \; v_2)$

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$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \qquad \lambda_1 = 2 \qquad \lambda_2 = 4$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$T = (\underline{v^1} \ \underline{v^2}) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$Av = \lambda v$$

Λ ?

$$\Lambda = T^{-1} A T$$

$$Av^1 = \lambda_1 v^1 = 2v^1 + 0v^2$$

$$Av^2 = \lambda_2 v^2 = 0v^1 + 4v^2$$

$$\Lambda = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

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Diagonalizable systems

We say that a square matrix A *is* diagonalizable *if it is similar to a diagonal matrix, that is, there is an invertible matrix* T *and a diagonal matrix* Λ *such that*

$$\Lambda = T^{-1}AT.$$

對角化

對角矩陣

A is diagonalizable if and only if for each λ_i, GM = AM ⇒ nondefective matrix

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Diagonalizable systems

$$\Lambda = \begin{pmatrix} & & \\ & \ddots & \\ & & \end{pmatrix} = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_n \end{pmatrix}$$

Theorem 2 *Suppose that for each distinct eigenvalue of A, its geometric multiplicity is the same as its algebraic multiplicity . Then A is similar to a diagonal matrix Λ whose diagonal elements are the eigenvalues of A. (If an eigenvalue has multiplicity m, it appears m times on the diagonal of Λ)*

$$\Lambda = T^{-1} A T \qquad \lambda_i \in \mathbb{C}$$

proof

$$T = (v^1 \; v^2 \; \cdots \; v^n)$$

$$Av^i = \lambda_i v^i$$

$$\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix}$$

complex Jordan form of A

useful result $= \mathrm{diag}(\lambda_1, \lambda_2, \cdots, \lambda_n)$

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Diagonalizable continuous-time systems

Consider a continuous-time system

$$\dot{x} = Ax$$

and suppose A is an $n \times n$ diagonalizable matrix, that is, there is a nonsingular matrix T such that

$$\Lambda = T^{-1}AT$$

is diagonal. Hence, the diagonal elements of Λ are the eigenvalues (including possible multiplicities) $\lambda_1, \lambda_2, \ldots, \lambda_n$ of A. Introducing the state transformation,

$$x = T\xi$$

we have $\xi = T^{-1}x$ and the system behavior is now equivalently described by

$$\dot{\xi} = \Lambda\xi\,.$$

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$x = T\xi \quad x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \quad \xi = \begin{pmatrix} \xi_1 \\ \vdots \\ \xi_n \end{pmatrix}$

$\dot{x} = Ax \Rightarrow \dot{\xi} = \Lambda \xi$, A is diagonalizable

$\Lambda = \mathrm{diag}(\lambda_1, \cdots \lambda_n)$

$\dot{\xi}_1 = \lambda_1 \xi_1$

$\dot{\xi}_2 = \lambda_2 \xi_2$

$\vdots$

$\dot{\xi}_n = \lambda_n \xi_n$

$\xi_0 = \xi(0) = (\xi_{10}, \cdots, \xi_{n0})$

$\xi(t) = \begin{pmatrix} e^{\lambda_1 t} & & & \\ & e^{\lambda_2 t} & & \\ & & \ddots & \\ & & & e^{\lambda_n t} \end{pmatrix} \xi_0$

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$$x(t) = \xi_1(t)v^1 + \cdots + \xi_n(t)v^n$$

$$= e^{\lambda_1 t}\xi_{10}v^1 + \cdots + e^{\lambda_n t}\xi_{n0}v^n$$

$\Rightarrow$ Every motion (sol) of a diagonalizable LTI Σ is "a sum of modes"

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Diagonalizable continuous-time systems

State transition matrix. Since $x(t) = T\xi(t)$ and $\xi_0 = T^{-1}x_0$, it follows from (6.14) that

$$x(t) = \Phi(t)x_0 \tag{6.16}$$

where the state transition matrix $\Phi(\cdot)$ is given by

$\Lambda = \mathrm{diag}(\lambda_1, \cdots \lambda_n) = T^{-1}AT$

↳ $\mathrm{diag}(e^{\lambda_1 t}, \cdots e^{\lambda_n t})$ 當矩陣已對角化的 sys

$\Phi(t) = T \begin{pmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{pmatrix} T^{-1}$

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Diagonalizable continuous-time systems

Example 81 Consider a CT system with A from example 62, that is,

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

Here,

$$T = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

and $\lambda_1 = 2, \lambda_2 = 4$.

$\Lambda = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \Rightarrow$ sol $= \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{4t} \end{pmatrix}$

$\Phi(t) = T \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{4t} \end{pmatrix} T^{-1} = \frac{1}{2} \begin{pmatrix} e^{2t}+e^{4t} & e^{2t}-e^{4t} \\ e^{2t}-e^{4t} & e^{2t}+e^{4t} \end{pmatrix}$

$x(t) = \Phi(t) x_0$

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Diagonalizable DT systems

$$x(k+1) = Ax(k)$$

A is diagonalizable, there is a nonsingular T matrix, such that

$$\Lambda = T^{-1}AT$$

is diagonal, the diagonal elements are $\lambda_1, \ldots, \lambda_n$

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$x = T\xi, \quad \xi = T^{-1}x$

$\xi(k+1) = \Lambda \xi(k)$

$\xi_1(k+1) = \lambda_1 \xi_1(k)$

$\vdots$

$\xi_n(k+1) = \lambda_n \xi_n(k)$

$\xi(0) = \xi_0$

$$\xi(k) = \begin{pmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{pmatrix} \xi_0$$

$x(k) = \Phi(k) x(k+1)$

$$\Phi(k) = T \begin{pmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{pmatrix} T^{-1}$$

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Example 82 Consider a DT system with A from example 62, that is,

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

Here,

$$T = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

and $\lambda_1 = 2, \lambda_2 = 4$.

$\xi = Tx$

$\xi(k) = \begin{pmatrix} 2^k & 0 \\ 0 & 4^k \end{pmatrix} \xi_0$

AT

$\Lambda = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \rightarrow SOS^{-1} = \begin{pmatrix} 2^k & 0 \\ 0 & 4^k \end{pmatrix}$

$\Phi(k) = T \begin{pmatrix} 2^k & 0 \\ 0 & 4^k \end{pmatrix} T^{-1}$

$= \frac{1}{2} \begin{pmatrix} 2^k + 4^k & 2^k - 4^k \\ 2^k - 4^k & 2^k + 4^k \end{pmatrix}$

$x(k) = \Phi(k) x_0$

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Real Jordan form for real diagonalizable systems

Suppose A is real and has a complex eigenvalue

$$\lambda = \alpha + j\omega$$

where α, ω are real. Then any corresponding eigenvector v is complex and can be expressed as

$$v = u + jw$$

where u, w are real and are linearly independent over the real scalars.

$Av = \lambda v$

$Au = \alpha u - \omega p$

$Ap = \omega u + \alpha p$

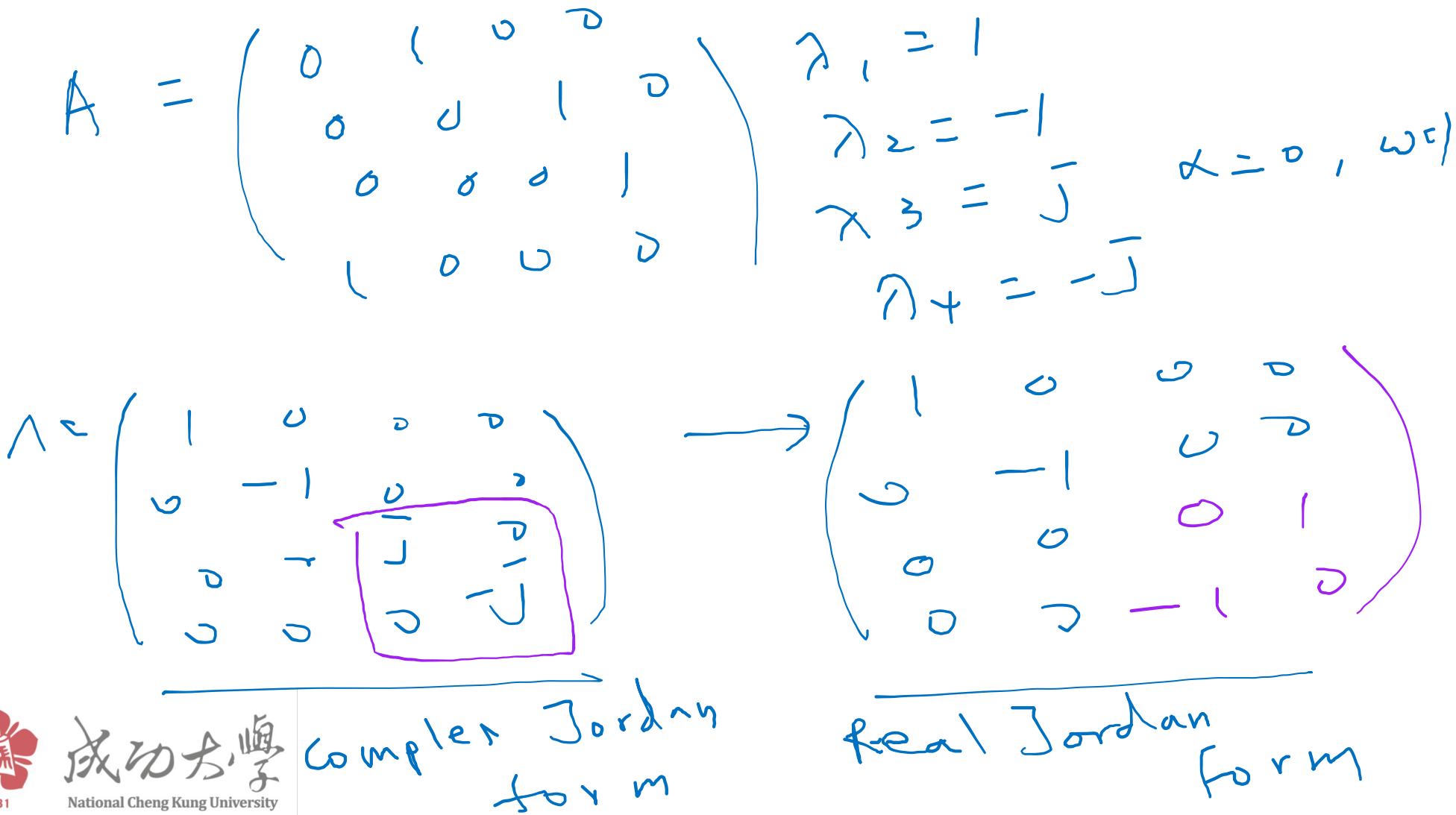
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$$\Lambda = \operatorname{diag}\left(\lambda_1, \lambda_2, \cdots \begin{pmatrix} \alpha_i & \omega_i \\ -\omega_i & \alpha_i \end{pmatrix} \cdots \lambda_n\right)$$

$$= \begin{pmatrix} \lambda_1 & & & & & \\ & \lambda_2 & & & & \\ & & \ddots & & & \\ & & & \begin{pmatrix} \alpha_i & \omega_i \\ -\omega_i & \alpha_i \end{pmatrix} & & \\ & & & & \ddots & \\ & & & & & \lambda_n \end{pmatrix}$$

real Jordan form

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Continuous Time System

The solutions of

$$\dot{x} = Ax, \qquad x(0) = x_0$$

are given by

$$x(t) = \Phi(t)x_0$$

where the state transition matrix $\Phi(\cdot)$ is given by

$$\Phi(t) = T\hat{\Phi}(t)T^{-1}$$

with $\hat{\Phi}(t)$ being the block diagonal matrix:

$$\text{block-diag}\left(e^{\lambda_1 t}, \ldots, e^{\lambda_l t}, \begin{pmatrix} e^{\alpha_1 t}\cos(\omega_1 t) & e^{\alpha_1 t}\sin(\omega_1 t) \\ -e^{\alpha_1 t}\sin(\omega_1 t) & e^{\alpha_1 t}\cos(\omega_1 t) \end{pmatrix}, \ldots, \begin{pmatrix} e^{\alpha_m t}\cos(\omega_m t) & e^{\alpha_m t}\sin(\omega_m t) \\ -e^{\alpha_m t}\sin(\omega_m t) & e^{\alpha_m t}\cos(\omega_m t) \end{pmatrix}\right)$$

T (Real Jordan form sol) T^{-1}

$\Lambda = (\lambda_1, \lambda_2, \begin{pmatrix} \alpha & \omega \\ -\omega & \alpha \end{pmatrix} \ldots \lambda_l)$

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State plane portraits for CT systems



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